

Representation Analysis of a Multi-Task Vision Transformer

Probes, CKA, and Sparse Feature Decomposition

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Project 1's finding. The same input produced different attribution maps depending on which of the three ViT variants was being explained.

Are the three variants actually different inside, or are the explanation methods just noisy?

- Project 1's attribution methods only describe outputs.
- To answer the question we have to look at the activations themselves.
- All four main methods of Project 2 produce numbers, not heatmaps. None overlap with Project 1.

Two Outcomes, Both Worth Reporting

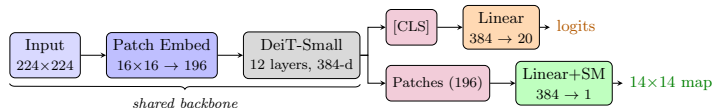
Outcome A: internal differences match attribution differences.

Project 1's attribution-map differences were real signal. The variants encode the input differently inside the network. Multi-task supervision genuinely changes what the backbone learns.

Outcome B: internal representations look the same. Project 1's differences were noise from the explanation methods. The disagreement was an XAI-reliability story rather than a model story.

Spoiler. The answer is partly Outcome A. Multi-task and classifier-only are close internally. Saliency-only diverges sharply.

Architecture: One Model, Three Loss Conditions



All three variants share this exact architecture. They differ only in which head receives gradient during fine-tuning.

- **cls_only**: BCE on classification head only.
- **sal_only**: KL on saliency head only.
- **multi-task**: both losses, $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{sal}}$, $\lambda = 1$.

No retraining for Project 2. We reuse the Project 1 checkpoints.

Data and Experimental Subsets

Dataset. SALICON (10k training and 5k validation fixation density maps from thousands of observers) paired with COCO 2014 top-20 category labels.

Each Project 2 method uses a fixed subset of the SALICON validation split:

Analysis	Subset size
Linear probing (per layer, 3 variants)	1000 images
CKA, Procrustes, SVCCA (per layer, 3 pairs)	1000 images
Patch-token clustering (final layer)	1000 images
SAE training and ablation	1000 train, 200 ablation
Grad-CAM and kernel SHAP	100 images

Activations are collected via PyTorch forward hooks on each transformer block. The same 1000-image subset is reused across probes, CKA, and clustering so the comparisons are paired.

Internal-Representation Methods: Why Each One

Each method answers a different sub-question about the backbone. Together they cover localization, similarity, geometry, and causal structure.

- **Linear probing.** *Where in the network does each task's information first become linearly decodable?* Probes answer this layer by layer. Nothing else does. Random-label control rules out probe capacity. (Alain & Bengio 2016, Hewitt & Liang 2019)
- **CKA, Procrustes, SVCCA.** *Are two variants encoding the input similarly?* Three measures on the same question. Pairing them defends against any single metric being confounded (next slide). (Kornblith 2019, Davari 2023)
- **Patch-token K-means clustering.** *What geometric structure do the embeddings form?* Probes and CKA describe content. Clustering describes shape. Are tokens organized by category, by saliency, both, or neither?
- **TopK sparse autoencoder** (exploratory). *Which directions in feature space causally drive each head?* The first three methods describe the representation. The SAE intervenes on it. (Stevens 2025, Paulo & Belrose 2025)

What CKA, Procrustes, and SVCCA Each Measure

All three compare two activation matrices $X, Y \in \mathbb{R}^{n \times d}$ (one per model variant). They differ in what kind of similarity they reward.

CKA (Kornblith et al. 2019): a normalized kernel-based independence statistic on centered Gram matrices.

$$\text{CKA}(X, Y) = \text{HSIC}(X, Y) / \sqrt{\text{HSIC}(X, X) \text{HSIC}(Y, Y)}$$

HSIC (Hilbert-Schmidt Independence Criterion, Gretton et al. 2005) is a kernel-based statistic that is zero exactly when X and Y are independent. Larger means more dependent. Linear CKA uses the inner-product kernel. RBF CKA uses $k(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2)$. Invariant to orthogonal rotation and isotropic scaling. Saturates near 1.0 between similar networks.

Procrustes (Schönemann 1966): align Y to X by an orthogonal matrix and report how well it fits. After centering and Frobenius-normalizing both matrices, the score is $\sum_i s_i$ where s_i are the singular values of $X^\top Y$. Rewards alignment achievable by one rotation. Less saturating than CKA.

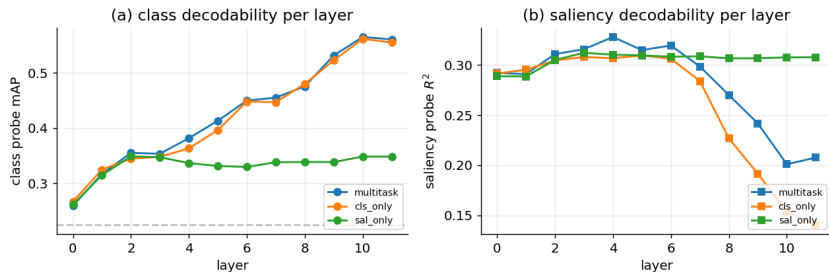
SVCCA (Raghu et al. 2017): take the top- k singular vectors of X and Y , then compute canonical correlations between those rank- k subspaces. Reports the mean correlation. Focuses on the dominant components and ignores low-variance noise. Depends on k (we use $k = 20$).

Required for the class-method benchmark. Both produce input heatmaps for visual comparison.

- **Grad-CAM** (class-method baseline, Lecture 9): representation-space attribution at block 10. Per-channel gradient weights times feature-map activations, ReLU-thresholded. (Selvaraju 2017)
- **Kernel SHAP** (additional): patch-grid Shapley regression with 200 mask samples per image and the efficiency constraint $\sum_j \phi_j = f(\text{full}) - f(\text{empty})$. (Lundberg & Lee 2017)

Grad-CAM uses block 10 rather than the final block because patch-token gradients at block 11 are zero (only the CLS token feeds the classification head).

Probes: Three Different Decodability Profiles

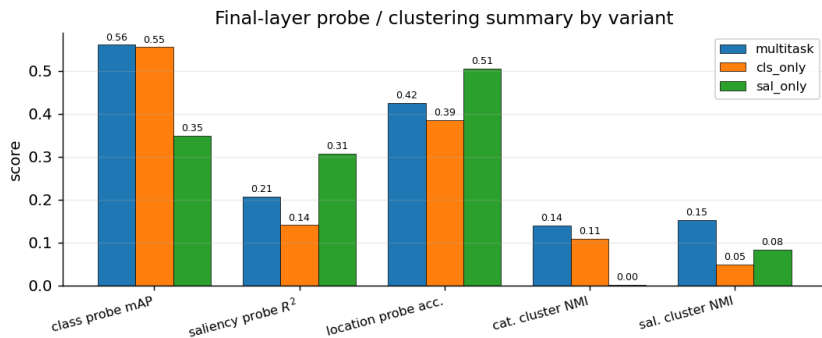


Higher is better on both panels. $\text{mAP} \in [0, 1]$ measures how well a linear probe can predict class labels from the layer's activations. $R^2 \in [-\infty, 1]$ measures how much of the saliency-map variance the probe explains.

- Class mAP at the final layer: **multi-task 0.561**, **cls-only 0.555**, **sal-only 0.348**.
- Saliency R^2 at the final layer: **sal-only 0.308**, **multi-task 0.208**, **cls-only 0.141**.

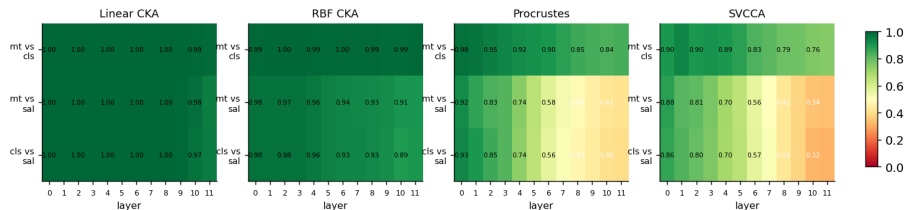
Multi-task gets full classification decodability and partial saliency decodability.

One-Slide Variant Summary



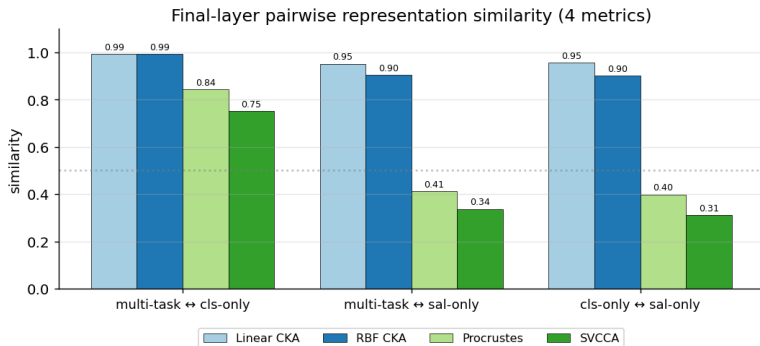
Higher is better on every metric. All five are scores in roughly $[0, 1]$, with 0 meaning random and higher values meaning the layer encodes more of that signal. Across all five, the multi-task variant matches or beats both single-task variants on every axis except the one each was trained for. It is the only variant whose patch-token clusters carry both category and saliency structure (NMI 0.140 and 0.153).

Pairwise Similarity, All 12 Layers



- Linear and RBF CKA stay green throughout: every pair, every layer, ≥ 0.90 . CKA cannot tell the variants apart.
- Procrustes and SVCCA reveal the asymmetry by mid-network. multi-task vs. cls-only stays warm-green. The two pairs involving sal-only fall into orange and red by layers 8 to 11.

Final-Layer Similarity, Side by Side



Higher = more similar (all four metrics are in $[0, 1]$, with 1 meaning identical activations and 0 meaning unrelated).

- multi-task vs. cls-only: Procrustes 0.84, SVCCA 0.75.
- multi-task vs. sal-only: Procrustes 0.41, SVCCA 0.34.
- cls-only vs. sal-only: Procrustes 0.40, SVCCA 0.31.

The multi-task backbone's geometry inherits from cls-only, not sal-only.

Why CKA Was Insufficient

The trap. Linear CKA gave essentially no signal: every variant pair scored ≥ 0.95 at every layer.

- Davari et al. (2023). CKA can be made arbitrarily high under subset-translation of the activation matrix. Saturation is documented.
- Cui et al. (2022). CKA is confounded by input-population structure, so even random networks can score high.
- Murphy et al. (2024). Even debiased CKA has limited resolution at low sample size.

Procrustes (orthogonal alignment after centering and Frobenius normalization) and **SVCCA** (canonical correlation between top- k singular subspaces) avoid both pitfalls and gave us the headline result.

Methodological recommendation. When comparing models that share initialization or training regime, never rely on CKA alone.

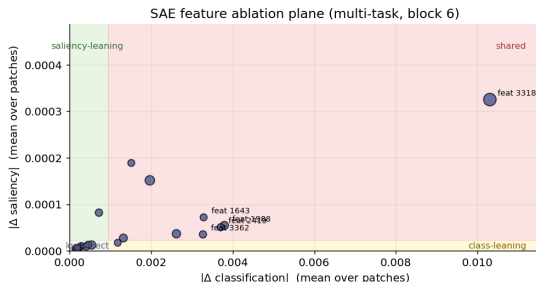
Patch-Token Clustering: Which Signal Each Variant Encodes

Variant	Cat. Purity	Cat. NMI	Sal. Purity	Sal. NMI	Stab. ARI
Multi-task	0.661	0.140	0.558	0.153	0.606
Classifier-only	0.644	0.109	0.442	0.049	0.649
Saliency-only	0.636	0.001	0.486	0.084	0.763

Metric definitions (all in $[0, 1]$, higher is better):

- **Purity**: fraction of tokens whose cluster's plurality label matches the token's true label. Sensitive to cluster-size imbalance.
- **NMI** (normalized mutual information): how much information cluster assignment shares with the true label, normalized so random clustering is 0 and perfect agreement is 1.
- **ARI** (adjusted Rand index): chance-corrected agreement between two clusterings. We use it for seed-to-seed stability (1 = identical clusters across seeds, 0 = chance).
- **Multi-task** is the only variant whose clusters carry both category and saliency structure.
- **sal-only** clusters carry essentially no class signal (NMI 0.001 is at the random floor).

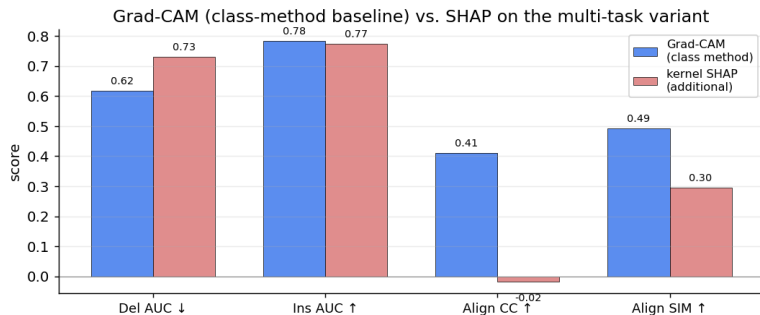
SAE Feature Ablation



An **SAE feature** is one of the 4096 sparse-dictionary directions the autoencoder learns. Each dot is one feature. We write Δcls for the average change in the classification head's output when that feature is zeroed out, and Δsal for the same change on the saliency head. Far right means class-driving. Far up means saliency-driving. Quadrants are split at the medians.

- Most features land class-leaning. Large Δcls , tiny Δsal .
- Top feature 3318 (active in 38% of patches): $\Delta \text{cls} = 0.010$, $\Delta \text{sal} = 0.0003$.
- No purely saliency-leaning feature at this $k=32$ sparsity.
- **Caveat.** TopK SAEs are seed-unstable (Paulo & Belrose 2025). Exploratory.

Attribution Baselines: Grad-CAM vs. Kernel SHAP



- Grad-CAM CC = 0.41. This is the highest of any method across both projects (Project 1 best was IG at 0.25).
- Kernel SHAP CC $\approx$ 0. No human alignment, even though Del and Ins AUC are comparable.
- **Pattern.** Representation-space methods (Grad-CAM, IG) align with human gaze. Perturbation-based methods (LIME, SHAP) do not.

Putting the Story Together

Going back to the central question. Are Project 1's attribution-map differences anchored in real internal differences?

Answer: yes, and asymmetrically.

- ➊ Multi-task and classifier-only are geometrically close in the second half of the network (Procrustes 0.84, SVCCA 0.75).
- ➋ Saliency-only is far from both (Procrustes ≈ 0.41 , SVCCA ≈ 0.33).
- ➌ Multi-task's clusters are the only ones simultaneously aligned with COCO categories and saliency quartiles.
- ➍ SAE feature ablation: class effects roughly $10\times$ saliency effects on the same heads.

Reading. Multi-task supervision adds saliency-aligned structure on top of a still-dominantly-classifier-aligned representation. It does not reorganize the backbone from scratch.

- **Single backbone, single dataset.** DeiT-Small and SALICON/COCO top-20 only. Larger ViTs and CNNs may behave differently.
- **SAE results are exploratory.** One layer (block 6), one seed, 4096-feature dictionary. Paulo & Belrose (2025) showed only $\sim 30\%$ of TopK features overlap across seeds, so any feature-inventory claim would need multi-seed work.
- **Kernel SHAP under sample budget.** 200 mask samples per image is below the standard for accurate Shapley estimates. Population CC ≈ 0 is robust to this. Per-image attributions are noisy.
- **Probing $n=1000$ on 384-dim activations** is small relative to dimensionality. Probe results would benefit from $n \geq 10,000$.

- **Cross-variant SAEs with shared dictionary** (Lindsey et al. 2024 crosscoder style). Train one SAE on activations of all three variants jointly to formally identify shared, classifier-leaning, and saliency-leaning features.
- **ERF-based feature interpretation** (Han, Kim, Kwak 2025) instead of activation-pattern interpretation, since attention mixing makes naive max-activating-patch readings misleading on ViTs.
- **Replicate the asymmetry on a CNN backbone.** Tests whether multi-task-inherits-from-classifier is architecture-specific or general.
- **Bigger probing n , more SAE seeds, factorial three-factor ablation** for a methodology-paper version.

Five Takeaways

- 1 The multi-task ViT is asymmetrically classifier-aligned. It is closer to cls-only than to sal-only by mid-network.
- 2 Linear CKA was uninformative. Procrustes and SVCCA carried the analysis. Use multiple similarity metrics whenever variants share initialization.
- 3 Multi-task's patch-token clusters are the only ones simultaneously aligned with COCO categories and saliency quartiles.
- 4 SAE feature ablation: class effects roughly $10\times$ saliency effects (single-seed caveat applies).
- 5 Grad-CAM achieved CC 0.41 against human saliency, the best alignment across both projects. Kernel SHAP achieved ≈ 0 , mirroring Project 1's LIME finding.

Questions?

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github.com/nickvpn/vit-ml-xai-project